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Review

Thermodynamic reassessment of the cerium–gold binary system

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ABSTRACT

Phase relationships in Ce–Au binary system have been thermodynamically assessed by using the CALPHAD technique through Thermo–Calc software package based on the evaluation of all available experimental data from the published literature. This data concern the phase equilibria and thermodynamic properties of the studied system. The intermetallic compounds Ce_2Au, Ce_9Au_{11}, $CeAu_2$, $Ce_{14}Au_{51}$ and CeAu6 were treated as stoichiometric phases. The two–sublattice energy model was employed to describe the CeAu intermetallic phase which presents a homogeneity range. The solution phases, including liquid, fcc_A1, hcp_A3 and bcc_A2 were treated as substitutional phases. Furthermore, the excess Gibbs energies of the studied phases were expressed with the Redlich–Kister polynomial functions. A set of self–consistent thermodynamic parameters formulating the Gibbs energy of various phases in the Ce–Au system were then obtained. A much better agreement was obtained between the calculated results and the reported experimental data.

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1. Introduction

Gold–Rare Earth Elements (REEs) alloys have been widely used during recent years in the fabrication of modern electronic devices, because of their great proprieties [1–3]. REEs are potential candidate to replace Au or at least to reduce its use in the eutectic Au–20 wt % Sn solders broadly used in electronics industry in order to avoid the scarce resources and high prices of Au as well as the poor mechanical properties of the Au–Sn intermetallic compounds.

Indeed, REEs can refine the mechanical properties and improve the wettability of solders [4] and are also able to enhance the fatigue performance under slow deformation [5–7] by reducing the oxidization of solder and by cleaning the grain boundaries.

The use of REEs in corrosion protection as inhibitors is very attractive because of their non-toxicity, as compared to the common chromate inhibitors [8]. A brief analysis of literature indicates that a protective effect can be provided by small amounts of REEs [8,9]. Furthermore, REEs are becoming increasingly important in the transition to a green economy, due to their crucial role in

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permanent magnets, lamp phosphors, catalysts and rechargeable batteries [10]. On the other hand, the field of starting materials for phosphor preparation, such mixed rare earth oxides, noted a strong evolution particularly in the area of fluorescent lighting devices [11].

The progress in the performance of catalytic systems dictated by the more and more severe regulations planned in Europe and USA to lower the level of pollutant emissions arise the cerium compounds as an interesting candidates owing to their redox properties [11]. For example CeO_2acts as a water gas shift and steam reforming promoter, diminishing the amount of CO and hydrocarbons through their oxidation in globally reducing conditions. Cerium oxide also stabilizes the dispersion of the precious metal particles, avoiding sintering, which would make them inactive [12,13]. It has been widely proved [14] that it is possible to prepare gold nanoparticles deposited on metal oxide supports [15], which exhibit high catalytic activity towards oxidation reactions [16–25]. In addition, gold supported on cerium oxide has been shown to possess high activity for volatile organic compounds (VOC) oxidation [26]. Cerium has been widely used in catalysis to purify vehicle exhausts and becomes the most rare earth oxide for controlling pollutant emission.

The imbalance in the REEs supply and demand has increased the importance of recycling and extraction of REEs. An efficient recycling route has already been developed to separate REEs and noble metals from many electronic applications which have complicated architectures and often contain very small quantities of those alloys [27,28].

An important step in investigating alloy systems is the definition of their phase diagrams. Knowledge of phase equilibria, phase stability, phase transformations are important reference points in the description and understanding both of the fundamental proprieties of the alloys and of their possible technological applications. Hence, the purpose of the current work is to assess the Ce–Au binary system for which the experimental phase diagram was recently reexamined by Lomello–Tafin et al. [29] and to develop a precise thermodynamic description by means of the CALPHAD technique (CALculation of PHAse Diagram) described and presented by Refs. [30,31] and successfully applied in various computer programs in order to give a more complete and consistent description of the phases existing in the studied system and their stability. The thermodynamic optimization of the Ce–Au system is part of a research program, carried out in our laboratory, and it focuses on the study of the interaction between rare earth elements and the noble metals.

2. Review of experimental data

The first version of the Ce–Au phase diagram was experimentally determined by Vogel and Heumann [32] who reported the existence of four compounds: Ce_2Au, CeAu, $CeAu_2$ and $CeAu_3$ as well as the invariant reactions involved by these phases. Mcmasters et al. [33] investigated the crystal structures of the Ce–Au intermetallic compounds, particularly in the rich gold part, except for $CeAu_6$. They found that the previously accepted $CeAu_3$ phase turned out to be a $CeAu_{3.6}$ compound confirming the already experimental results of Donolato and Steeb [34]. By means of X–Ray Diffraction (XRD), Gschneidner et al. [35–37] reinvestigated the crystal structure of $CeAu_{3.6}$ and adopted the $Ce_{14}Au_{51}$ formulation for this phase. Concerning the equiatomic CeAu phase, Mcmasters et al. [33] gave the melting point of βCeAu, and the temperature of the αCeAu ↔ βCeAu allotropic transition. Moreau and Parthé [38] reexamined the crystal structure of CeAu6 and found that this phase is isostructural with PrAu6, which has a monoclinic structure. Later, Ferro et al. [39] observed the existence of a new Ce3Au4 phase with a hexagonal Pu3Au4–type structure. However, in a new version of the Ce–Au phase diagram redrawn by Okamoto [40], the Ce3Au4compound was represented by a dashed line, because no evidence of the existence of this phase was found in the ternary Ce–Sn–Au isothermal section studied by Boulet et al. [41] at 1023 K, and also because no crystal structure data relative to the Ce3Au4 phase were found in the Pearson's Crystal Database [42]. For this reasons, we have not considered this phase in the current optimization, in contrast with the previous work by Dong et al. [43]. Most recently, Lomello–Tafin et al. [29] have reconsidered the Ce–Au system (Fig. 1) using XRD, Differential Thermal Analysis (DTA), Optical and Scanning Electron Microscopy (SEM), Energy–Dispersive X–ray Spectroscopy (EDS) and confirmed the presence of the previously discovered phases except the Ce3Au4 phase for which these authors adopted the Ce9Au11 formula, even if the XRD patterns was indexed in the hexagonal Pu3Pd4 structure type [29]. The melting point of the CeAu2 phase, measured by Ref. [29] was 1479 K, higher by 76 K than the determined one by Vogel et al. [32]. According to [29] the difference may be due to purity of the Cerium used by Ref. [32]. Furthermore, these authors discover the existence of the δCe ↔ γCe + liquid metatectic reaction which occurs at 969 ± 3 K and established the existence of a stability domain for the βCeAu phase which extends up to 52 at.% Au at 1470 K. Lomello–Tafin et al. [29] reported also the existence of a $\alpha Ce_{14}Au_{51} \leftrightarrow \beta Ce_{14}Au_{51}$ transition at 1203 K. The extensive experimental work of [29] gave also the temperatures and compositions of the Ce–Au invariant reactions (Table 4).

The crystal structures of the various phases of the Ce–Au system are shown in Table 1.

Regarding the thermochemical information of the Ce–Au system, only a limited amount of data can be found in literature. The first calorimetric measurements were carried out by Fitzner and Kleppa [44] who reported the standard molar enthalpies of formation of the CeAu, $CeAu_2$ and $Ce_{14}Au_{51}$ phases as well as the enthalpies of mixing of the liquid phase. Later, Lomello–Tafin et al. [29] measured by means of direct reaction calorimetry at high temperature, the standard enthalpies of formation of the CeAu and $CeAu_2$ phases. Using the same technique, Ferro et al. [39] determined the enthalpy of formation of the Ce_2Au phase at high temperature. Most recently, Janghorban et al. [45] measured at 1123 K, the enthalpies of formation of the CeAu, $CeAu_2$ and $Ce_{14}Au_{51}$ congruent compounds by means of direct reaction calorimetry and deduced their standard enthalpies of formation from the relative enthalpies increment of these phases measured

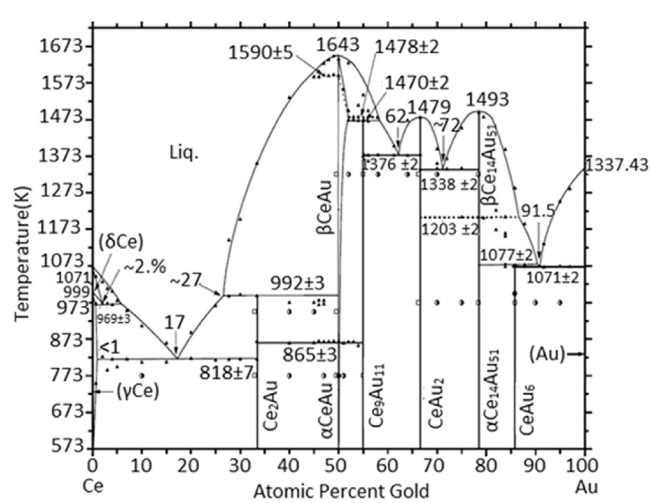

Fig. 1. Experimental Ce–Au phase diagram from Lomello–Tafin et al. [29].

Table 1
Crystal structures of the various phases in the Ce–Au system.

Phase	Composition at.% Au	Pearson symbol	Space group	Symbol used in thermo-calc data file	Prototype	Lattice Parameters (nm)			Ref.
						a	b	c	
Au	100[a]	cF4	$Fm\overline{3}m$	fcc_*A1*	Cu	0.40784[f]	–	–	[35,36]
αCe	0[b]	cF4	$Fm\overline{3}m$	fcc_*A1*	Cu	0.485[f]	–	–	[29,80]
βCe	0[b]	hP4	$P6_3/mmc$	hcp_A3	αLa	0.36810[f]	–	1.1857[f]	[29,80]
γCe	0–<1[c]	cF4	$Fm\overline{3}m$	fcc_A1	Cu	0.51610[f]	–	–	[29,81]
δCe	0–2[d]	cI2	$Im\overline{3}m$	bcc_A2	W	0.412[f]	–	–	[29,81]
Ce_2Au	33.4	oP12	Pnma	Ce_2Au	Co_2Si	0.7271	0.5073	0.9320	[33,73]
βCeAu	50–52[e]	oC8	Cmcm	βCeAu	CrB	0.3900[g]	1.114[g]	0.475[g]	[33,73]
αCeAu	50	oP8	Pnma	αCeAu	FeB	0.7430	0.466	0.593	[33,73]
Ce_9Au_{11}	55	hR42	-	Ce_9Au_{11}	Pu_3Pd_4	–	–	–	[29]
$CeAu_2$	66.7	oI12	Imma	$CeAu_2$	$CeCu_2$	0.4528	0.7203	0.8068	[72,73]
$\alpha Ce_{14}Au_{51}$	78.5	hP68	P6/m	$\alpha Ce_{14}Au_{51}$	$Ag_{51}Gd_{14}$	1.281 1.277		0.9327 ± 0.3 0.929	[33,73]
$\beta Ce_{14}Au_{51}$	78.5	–	–	$\beta Ce_{14}Au_{51}$	–	–	–	–	[29]
$CeAu_6$	85.7	mC28	C2/c	$CeAu_6$	$PrAu_6$	0.78 ± 1	0.778 ± 1	0.908 ± 1	[33,38,73]

[a] No range limits was reported in the literature.
[b] The solubility of Au in αCe and βCe phases was reported to negligible.
[c] Maximum of solubility reached at 818 k.
[d] Maximum of solubility reached at 969 k.
[e] Maximum of solubility reached at 1470 k.
[f] Lattice parameters for pure elements.
[g] Lattice parameters at stoichiometric composition.

from 298.15 K to 1123 K. We note that the experimental values reported by Fitzner and Kleppa [44] are less exothermic than those measured by Janghorban et al. [45] who explained that this difference cannot be due to the oxidation of cerium powder in the presence of residual oxygen during experiments, but can be attributed to the difference in the sample preparation technique. Due to the insufficiency of thermodynamic data, the ab initio calculation method [46–49] was employed by Dong et al. [43] to calculate the enthalpies of formation of intermetallic compounds in the course of a thermodynamic assessment of the Ce–Au system. We note that the ab initio calculated values are close to the calorimetric ones by Janghorban et al. [45]. In Table 2, we summarize the different enthalpies of formation values obtained by various authors and the ones calculated in the current work.

We note that the discrepancies in the published experimental results are related to the difficulties encountered by several authors

Table 2
Comparison between measured and optimized enthalpies and entropies of formation of the Ce–Au intermetallic compounds.

Intermetallic compounds	Enthalpy of formation ΔH_f° (kJ/mol.at.)	Technique used	T (K)	Entropy of formation ΔS_f° (J/mol.K)	Reference.
Ce_2Au	−55 ± 1.0	Calorimetry	298.15	–	[39]
	−50.99	ab initio	298.15	–	[46–49]
	−59.39	Optimization	298.15	−7.3464	[43]
	−61.0	Optimization	298.15	−4.6	This work
αCeAu	−74.0 ± 1.0	Calorimetry	1123	–	[29]
	−75.36	ab initio	298.15	–	[46–49]
	−75.2 ± 1.4	Calorimetry	1123	–	[45]
	−76.2 ± 1.9	Calorimetry	298.15	–	[45]
	−62.8 ± 4.2	Calorimetry	298.15	–	[44]
	−76.02	Optimization	298.15	−1.23	[43]
	−78.6	Optimization	298.15	−1.98	This work
βCeAu	−74.71	ab initio	298.15	–	[46–49]
	−75.70	Optimization	298.15	−0.81	[43]
	−79.0	Optimization	298.15	−2.43	This work
$CeAu_2$	−71.5 ± 2	Calorimetry	1123	–	[45]
	−73.0	Calorimetry	1123	–	[29]
	−57.5 ± 4.7	Calorimetry	298.15	–	[44]
	−71.46	ab initio	298.15	–	[46–49]
	−70.6 ± 2	Calorimetry	298.15	–	[45]
	−70.0	Optimization	298.15	−0.28	[43]
	−66.0	Optimization	298.15	−0.97	This work
$\alpha Ce_{14}Au_{51}$	−55.0 ± 1.7	Calorimetry	1123	–	[45]
	−44.0 ± 5	Calorimetry	1223	–	[29]
	−53.2 ± 1.9	Calorimetry	298.15	–	[45]
	−49.5 ± 2.2	Calorimetry	298.15	–	[44]
	−57.71	Optimization	298.15	−0.66	[43]
	−55.99	Optimization	298.15	−3.1	This work
$\beta Ce_{14}Au_{51}$	−53.1	Optimization	298.15	−5.4	This work
Ce_9Au_{11}	−73.82[a]	Optimization	298.15	−1.145	[43]
	−75.9	Optimization	298.15	−2.07	This work
$CeAu_6$	−39.76	ab initio	298.15	–	[46–49]
	−41.59	Optimization	298.15	−2.65	[43]
	−45.8	Optimization	298.15	−11.6	This work

[a] Calculated *value for the Ce_3Au_4 stoichiometry.*

Table 3
Optimized thermodynamic parameters for the Ce–Au system.

Phase	Thermodynamic models	Parameters in SI
Liquid	(Ce, Au)	${}^{0}L^{Liquid} = -269142 + 13.43*T$ ${}^{1}L^{Liquid} = -114055 + 73.21*T$ ${}^{2}L^{Liquid} = 49931.9 - 4.78*T$
bcc_A2	$(Ce, Au)_1:(Va)_3$	${}^{0}L^{bcc_\lambda 2} = -165689 + 1.5679*T$
fccA_1	$(Ce, Au)_1:(Va)_1$	${}^{0}L^{fcc_\lambda 1} = -152935$
Ce_2Au	$(Ce)_{0.667}:(Au)_{0.333}$	$G^{Ce2Au}_{CeAu} - 0.667*H^{fcc_A1}_{Au} - 0.333*H^{fcc_A1}_{Au} = 0.667G^{fcc_A1}_{Au} + 0.333G^{fcc_A1}_{Au} - 61052 + 4.56*T$
$CeAu_2$	$(Ce)_{0.333}:(Au)_{0.667}$	$G^{CeAu2}_{CeAu} - 0.333*H^{fcc_A1}_{Au} - 0.667*H^{fcc_A1}_{Au} = 0.333G^{fcc_A1}_{Au} + 0.667G^{fcc_A1}_{Au} - 66000 + 0.97*T$
βCeAu	$(Au)_{0.500}:(Au:Ce)_{0.500}$	$G^{\beta CeAu}_{CeAu} - H^{fcc_A1}_{Au} = G^{fcc_A1}_{Au} + 5000\,G^{\beta CeAu}_{Ce:Au} - 0.500*H^{fcc_A1}_{Ce} - 0.500*H^{fcc_A1}_{Ce} = 0.500*G^{fcc_A1}_{Ce} + 0.500*G^{fcc_A1}_{Ce} - 78995 + 2.432*T$ ${}^{0}L^{\beta CeAu}_{Au:Au,Ce} = -44800 + 7.5*T$
αCeAu	$(Au)_{0.500}:(Au:Ce)_{0.500}$	$G^{\alpha CeAu}_{Au:Au} - H^{fcc_A1}_{Au} = G^{fcc_A1}_{Au} + 5000G^{\alpha CeAu}_{Ce:Au} - 0.500*H^{fcc_A1}_{Ce} - 0.500*H^{fcc_A1}_{Ce} = 0.500*G^{fcc_A1}_{Ce} + 0.500*G^{fcc_A1}_{Ce} - 78600 + 1.98*T$ ${}^{0}L^{\alpha CeAu}_{Au:Au,Ce} = -45432 + 8.15*T$
Ce_9Au_{11}	$(Ce)_{0.450}:(Au)_{0.550}$	$G^{Ce9Au11}_{CeAu} - 0.450*H^{fcc_A1}_{Ce} - 0.550*H^{fcc_A1}_{Au} = 0.450*G^{fcc_A1}_{Ce} + 0.550*G^{fcc_A1}_{Au} - 75909 + 2.07*T$
$\beta Ce_{14}Au_{51}$	$(Ce)_{0.215}:(Au)_{0.785}$	$G^{\beta Ce14Au51}_{Ce:Au} - 0.215*H^{fcc_A1}_{Ce} - 0.785*H^{fcc_A1}_{Au} = 0.215*G^{fcc_A1}_{Ce} + 0.785*G^{fcc_A1}_{Au} - 53118 + 3.09*T$
$\alpha Ce_{14}Au_{51}$	$(Ce)_{0.215}:(Au)_{0.785}$	$G^{\alpha Ce14Au51}_{CeAu} - 0.215*H^{fcc_A1}_{Ce} - 0.785*H^{fcc_A1}_{Au} = 0.215G^{fcc_A1}_{Ce} + 0.785G^{fcc_A1}_{Au} - 55924 + 5.43*T$
$CeAu_6$	$(Ce)_{0.143}:(Au)_{0.857}$	$G^{CeAu6}_{CeAu} - 0.143*H^{fcc_A1}_{Ce} - 0.857*H^{fcc_A1}_{Au} = 0.143*G^{fcc_A1}_{Ce} + 0.857*G^{fcc_A1}_{Au} - 45844 + 11.64*T$

[50,51] in the preparation and study of rare earth alloys. Among these difficulties we cite:

- High reactivity with oxygen and several crucible materials, especially at high temperatures
- Problems with the purity of the metals and the difficulty of obtaining really pure rare earth metals and the strong effects of several common impurities in changing their constitutional properties.
- High melting temperatures of a number of rare earth metals and alloys.
- High vapour pressure and volatility of some rare earth metals.

3. Thermodynamic models

3.1. Pure elements

The Gibbs energy function $G_i^{\Phi}(T) = {}^{0}G_i^{\Phi} - H_i^{SER}$ for the element i (i = Ce, Au) in the phase Φ (Φ = Liquid, hcp_A3, fcc_A1 and bcc_A2) is described by an equation of the following form:

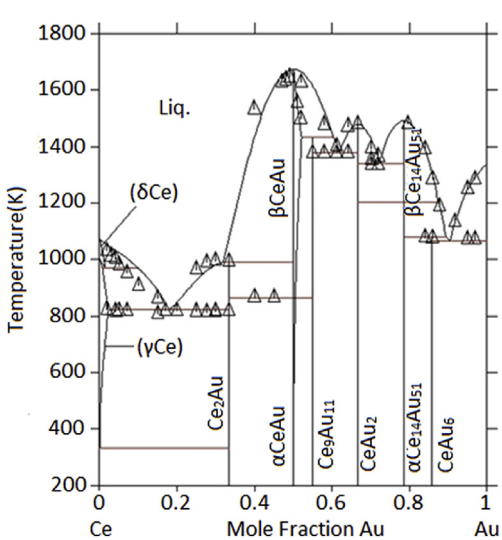

Fig. 2. Calculated Ce–Au phase diagram compared with the experimental data [29].

$$G_i^{\Phi}(T) = a + bT + cT + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9} \quad (1)$$

Where $H_i^{SER}(298.15K)$ is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, fcc_A1 for both Ce and Au.

In this work, the Gibbs energy functions are taken from the Scientific Group Thermo data Europe (SGTE) compiled by Dinsdale [52].

3.2. Solution phases

The liquid and the (γCe), (δCe) phases were modelled as substitutional solutions. The Gibbs energy of one mol of formula unit of phase Φ is expressed as the sum of the reference part ${}^{ref}G^{\Phi}$ the ideal part ${}^{id}G^{\Phi}$ and the excess part ${}^{exc}G^{\Phi}$:

$$G_m^{\Phi} = {}^{ref}G^{\Phi} + {}^{id}G^{\Phi} + {}^{exc}G^{\Phi} \quad (2)$$

$${}^{ref}G^{\Phi}(T) = \left({}^{0}G^{\Phi}_{Ce}(T) - H^{SER}_{Ce}(298.15K)\right)_{X_{Ce}} + \left({}^{0}G^{\Phi}_{Au}(T) - H^{SER}_{Au}(298.15K)\right)_{X_{Au}} \quad (3)$$

$${}^{id}G^{\Phi} = RT(X_{Ce}\ln X_{Ce} + X_{Au}\ln X_{Au}) \quad (4)$$

The excess terms of all the solution phases were modelled by the Redlich–Kister [53] formula:

$${}^{exc}G^{\Phi}(T) = {}^{X}Ce^{X}Au\left[\sum_{i=0}^{n} {}^{i}L^{\Phi}_{Ce,Au}(T)(X_{Ce} - X_{Au})^{i}\right] \quad (5)$$

$${}^{i}L^{\Phi}_{Ce,Au}(T) = a_i + b_iT \quad (6)$$

where ${}^{i}L^{\Phi}_{Ce,Au}(T)$ is the interaction parameter between the elements Ce and Au, which is evaluated in this study, T is the temperature in Kelvin, R is the gas constant and the coefficients a_i and b_i are optimized using the PARROT module [54], while x_{Ce} and x_{Au} are the mole fraction of Ce and Au elements respectively.

3.3. Stoichiometric compounds

The Gibbs energy of the stoichiometric compounds Ce_pAu_q is

Table 4
Comparison between experimental and calculated data relative to the Ce–Au invariant reactions.

Reaction	Data from current work				Literature data.				
	Compositions (at. % Au)			T(K)	Compositions (at. % Au)			T(K)	Ref
Liq ↔ Ce_2Au + γCe	18.1	33.4	2.31	821	16*	33.4	–	793	[32]
					16*	33.4	–	793*	[35]
					16	33.4	–	793	[36]
					16*	33.4	–	793*	[37]
					16.4	33.4	–	793	[43]
					17	33.4	<1	818 ± 7	[29]
Liq + βCeAu ↔ Ce_2Au	32	50	33.3	991	21.5*	50	33.3	930	[32]
					23.5*	50	33.3	938	[35]
					25.2	50	33.3	938	[36]
					24*	50	33.3	938	[37]
					27.7	50	33.3	940	[43]
					27.0	50	33.3	992 ± 3	[29]
Liq ↔ βCeAu	50			1645	50			1623	[32]
								1645 ± 10	[33]
								1645 ± 10	[35]
								1645	[36]
								1645 ± 10	[37]
								1645	[43]
								1643	[29]
αCeAu ↔ βCeAu	50			865	50			753 ± 5	[33]
								753 ± 10	[35]
								753 ± 10	[37]
								753	[43]
								865 ± 3	[29]
Liq ↔ $CeAu_2$ + Ce_9Au_{11}	62	66.7	55	1376	62	66.7	55	1376 ± 2	[29]
Liq ↔ $CeAu_2$	66.7			1477	66.7			1403	[32]
					66.7			1403	[35]
					66.7			1403	[36]
					66.7			1403	[37]
					66.7			1403	[43]
					66.7			1479	[29]
Liq ↔ $CeAu_2$ + $\beta Ce_{14}Au_{51}$	72.3			1341	69.0*			1423*	[32]
					69.0*			1423	[35]
					72.0			1423	[36]
					69.0*			1423	[37]
					72.4			1423	[43]
Liq ↔ $\beta Ce_{14}Au_{51}$	78.5			1491	75.0			1423	[32]
					78.5			1423	[35]
					78.5			1423	[36]
					78.5			1423	[37]
					78.5			1423	[43]
					78.5			1493	[29]
$\alpha Ce_{14}Au_{51}$ ↔ $\beta Ce_{14}Au_{51}$	78.5			1203	78.5			1203 ± 4	[29]
Liq + $\alpha Ce_{14}Au_{51}$ ↔ $CeAu_6$	89.5	78.5	85.7	1079	89*	78.5	85.7	1143*	[35]
					87.4	78.5	85.7	1143	[36]
					89*	78.5	85.7	1173	[37]
					_	78.5	85.7	<1023	[41]
					89.2	78.5	85.7	1021	[43]
					89.0	78.5	85.7	1077	[29]
Liq ↔ $CeAu_6$ + Au (fcc)	90.5	85.7	100	1065	90.2*	85.7	100	1081	[32]
					90.5*	85.7	100	1081*	[35]
					91.0	85.7	100	1081	[36]
					90.5*	85.7	100	1081	[37]
					_	85.7	100	<1023	[41]
					91.7	85.7	100	941	[43]
					91.5	85.7	100	1071 ± 2	[29]
Liq + βCeAu ↔ Ce_9Au_{11}	59.0	51.6	55.5	1470	58.0	52.0	55.5	1470 ± 2	[29]
δ(Ce) ↔ γ(Ce) + Liq	1.50	0.74	9.7	971	2	1	5.9	969 ± 3	[29]

* approximate value.

expressed as follows:

$$^0G_{Ce_pAu_q} = \frac{p}{p+q}\,^0G_{Ce} + \frac{q}{p+q}\,^0G_{Au} + a + bT \quad (7)$$

where $^0G_{Ce}$ and $^0G_{Au}$ are the Gibbs energy of the pure elements Ce and Au respectively, the parameters *a* and *b* were evaluated in the present work using the available standard enthalpies of formation in the literature.

3.4. Phase with homogeneity range

The CeAu phase (isotypic with FeB) which was reported to have an experimental homogeneity range according to [29] was treated by a two–sublattice model [55]. The experimental extension of CeAu is towards the Au rich part, so the modeling was as the formula $(Au)_{0.5}(Au, Ce\%)_{0.5}$ where the first sublattice is mainly occupied by Au and the second by Au and Ce elements. The symbol %

Table 5
Comparison between experimental and calculated compositions of the liquidus in the RE rich side of the equiatomic RE–Au compound.

Invariant reaction	Current work		Literature data		Ref.
	at. % Au	T/K	at. % Au	T/K	
Liq + βCeAu ↔ Ce_2Au	32	991	21.5[a]	930[a]	[32]
			27.7[b]	940[b]	[43]
			27[a]	992[a]	[29]
Liq + βNdAu ↔ Nd_2Au			30[a]	1083[a]	[79]
			32*	1085*	Work in progress
Liq + βSmAu ↔ Sm_2Au			22[a]	1138[a]	[74]
			31*	1136*	Work in progress
Liq + βTbAu ↔ Tb_2Au			29[a]	1273[a]	[75]
			32*	1270*	Work in progress
Liq + βLaAu ↔ La_2Au	–	–	24.9[a]	938[a]	[76]
			24.2[b]	941[b]	[43]
Liq + αErAu ↔ Er_2Au	–	–	30[a]	1393[a]	[77]
			26.4[b]	1387[b]	[43]
Liq + βHoAu ↔ Ho_2Au	–	–	30[a]	1368[a]	[77]
			29.3[b]	1378[b]	[43]
Liq + αTmAu ↔ Tm_2Au	–	–	31[a]	1433[a]	[77]
			33.6[b]	1441[b]	[43]
Liq + βPrAu ↔ Pr_2Au	–	–	26[a]	1023[a]	[78]
			30[b]	1023[b]	[66]

* Work in progress.
[a] Experimental value.
[b] Calculated value.

denotes the major component in considered sublattice.

The Gibbs energy function per mole of the formula unit $(Au)_{0.5}(Au, Ce\ \%)_{0.5}$ is the following:

$$G^{0.5Au0.5Ce} - H^{SER}_{0.5Au0.5Ce} = y^1_{Au}y^2_{Au}\,{}^0G^{Au:Au} + y^1_{Au}y^2_{Ce}\,{}^0G^{Au:Ce} + RT\left[0.5^{*}y^1_{Au}\ln y^1_{Au} + 0.5^{*}\left(y^2_{Au}\ln y^2_{Au} + y^2_{Ce}\ln y^2_{Ce}\right)\right] + {}^{exc}G \tag{8}$$

$$H^{SER}_{0.5Au0.5Ce} = 0.5H^{SER}_{Ce} + 0.5H^{SER}_{Au} \tag{9}$$

where $y^1_{Au} = 1$ denotes the site fraction of Au in the first sublattice, y^2_{Ce} and y^2_{Au} the site fractions of Ce and Au in the second sublattice. ${}^0G^{Au:Au}$ represents the Gibbs free energy of the hypothetical compound $Au_{0.5}Au_{0.5}$ while ${}^0G^{Au:Ce}$ is the Gibbs energy of the $Ce_{0.5}Au_{0.5}$ stoichiometric compound. ${}^{exc}G$ is the excess Gibbs energy expressed by the following expression:

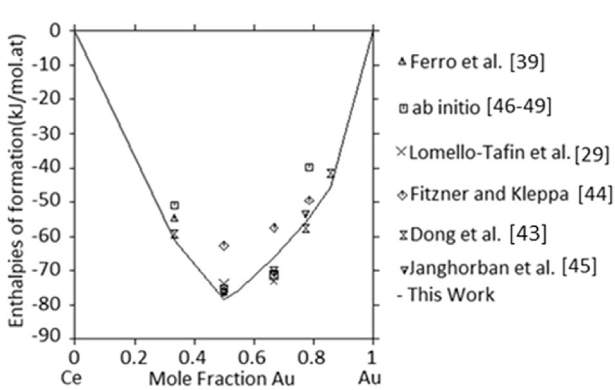

Fig. 3. Calculated standard enthalpies of formation of the Ce–Au intermetallic compounds with available literature data.

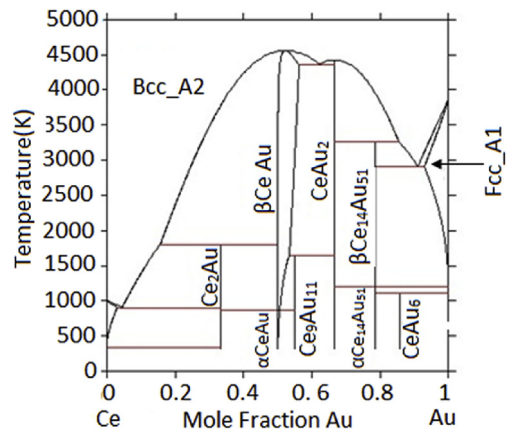

Fig. 4. Calculated Ce–Au phase diagram when the liquid phase is suspended.

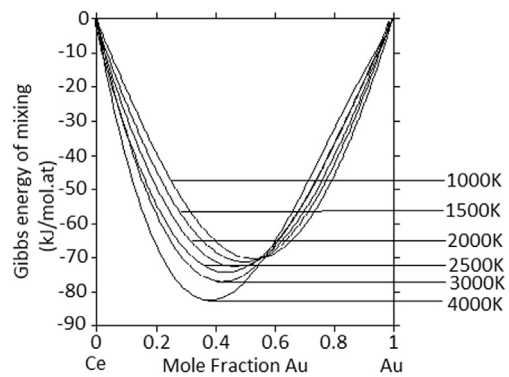

Fig. 5. Gibbs energy of mixing of the liquid phase at different temperatures. The reference states are liquid Ce and liquid Au.

$${}^{exc}G = y^2_{Au}y^2_{Ce}\left(y^2_{Au} - y^2_{Ce}\right)\left(L^{\lambda}Au : Au, Ce\right) \tag{10}$$

where $L^{\lambda}_{\lambda u:Au,Ce}$ $(\lambda = 0, 1, 2\ldots)$ represents the interaction parameters between the Ce and Au elements in the second sublattice while the first sublattice is only occupied by the Au element.

These excess parameters are temperature dependent as:

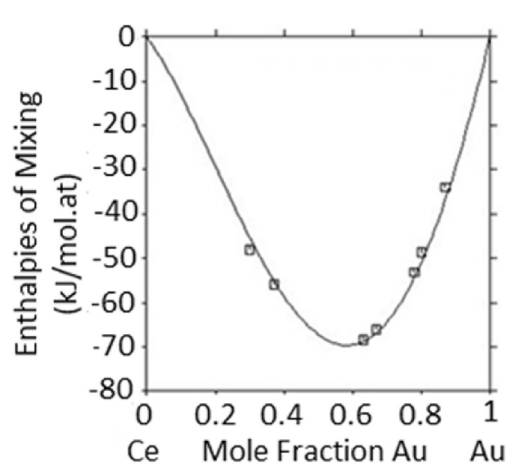

Fig. 6. Calculated enthalpies of mixing of the liquid phase compared with the calorimetric results by Fitzner and Kleppa [44].

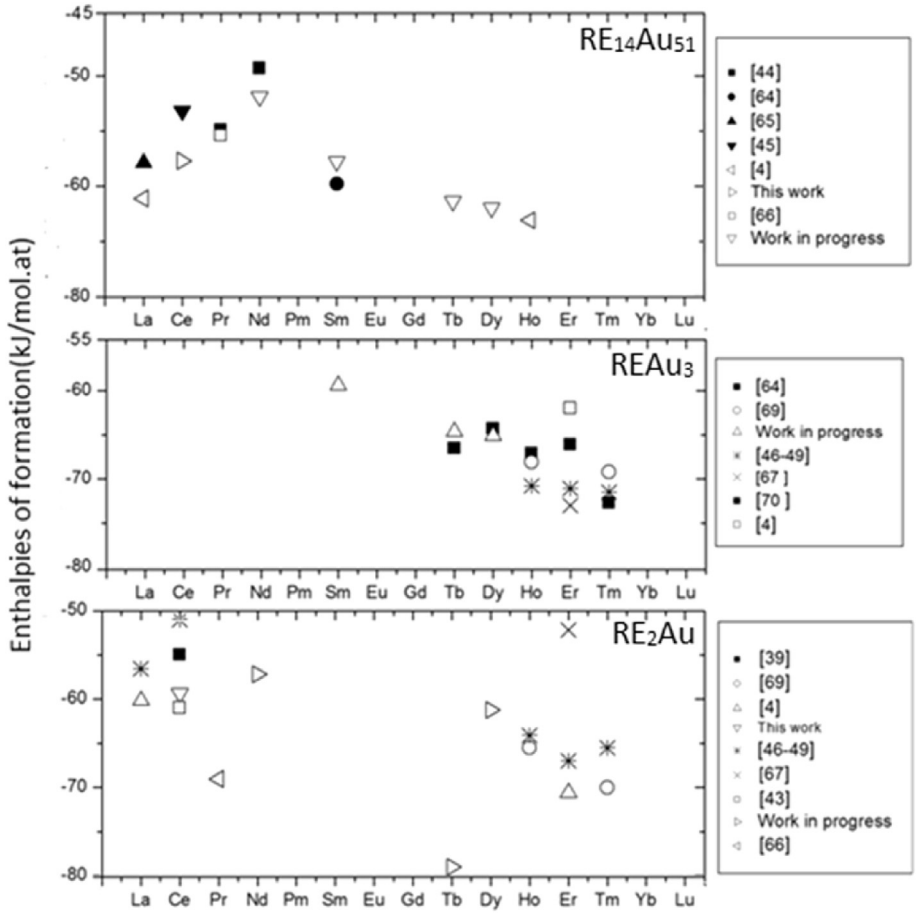

Fig. 7. Enthalpies of formation of the $RE_{14}Au_{51}$, $REAu_3$ and RE_2Au phases with data from work in progress (∇, Δ and ▷).

$$L^{\lambda}_{\lambda u:Au,Ce} = a_{\lambda} + b_{\lambda}T \quad (11)$$

a_{λ} and b_{λ} are the coefficients to be optimized.

In order to avoid the occurrence of the hypothetical compound $Au_{0.5}Au_{0.5}$ during the phase diagram calculation, a constant value of +5000 J/mol–atoms [56–58] was added to the GHSERAu function (see Table 3).

4. Optimization procedure

The thermodynamic parameters were evaluated using a recurrent runs of the PARROT module [54] in the Thermo–Calc software [59], which works by minimizing the square sum of the differences between experimental and computed values. In the assessment process, each piece of experimental information is given a certain weight according to its estimated accuracy. The weights were changed systematically during the assessment until most of the experimental data were accounted for within the claimed uncertainty limits.

The optimization was carried out by steps. The thermodynamic parameters for the liquid phase were first optimized using the experimental enthalpies of mixing of the liquid phase [44]. The second treatment was based on the available experimental standard enthalpies of formation for some Ce–Au intermetallic compounds and the phase boundaries information. During this treatment, all intermetallic phases were assumed to be stoichiometric compounds. In a third treatment, the CeAu phase was treated as a $(Au)_{0.5}(Au, Ce\ \%)_{0.5}$ formula using a two–sublattice model [55] in order to take into account its homogeneity range. The parameters obtained from the second treatment were used as starting values for the third treatment.

5. Results and discussion

The thermodynamic assessment of the Ce–Au system is carried out in the present work using the PARROT module in the Thermo–Calc software [59], taking into account all the relevant experimental results published by Lomello–Tafin et al. [29] and the recent experimental measurements obtained by Janghorban et al. [45] as well as the enthalpies of mixing of the liquid phase reported by Fitzner and Kleppa [44].

The Table 3 reports the thermodynamic parameters optimized for all condensed phases in the Ce–Au binary system obtained in the present work, based on the experimental data and thermodynamic proprieties available for the Ce–Au system. As discussed in the following sections, the present calculations can reproduce most of the recent experimental and calculated data available in the literature. The calculated compositions and temperatures of the invariant reactions in this binary system are summarized and confronted to the available experimental data in Table 4. The calculated invariant temperatures are in very good agreement with those determined experimentally by Lomello–Tafin et al. [29]. The difference is within the DTA experimental error reported, by these authors to be ±5 K.

A comparison between the phase diagram assessed in this work

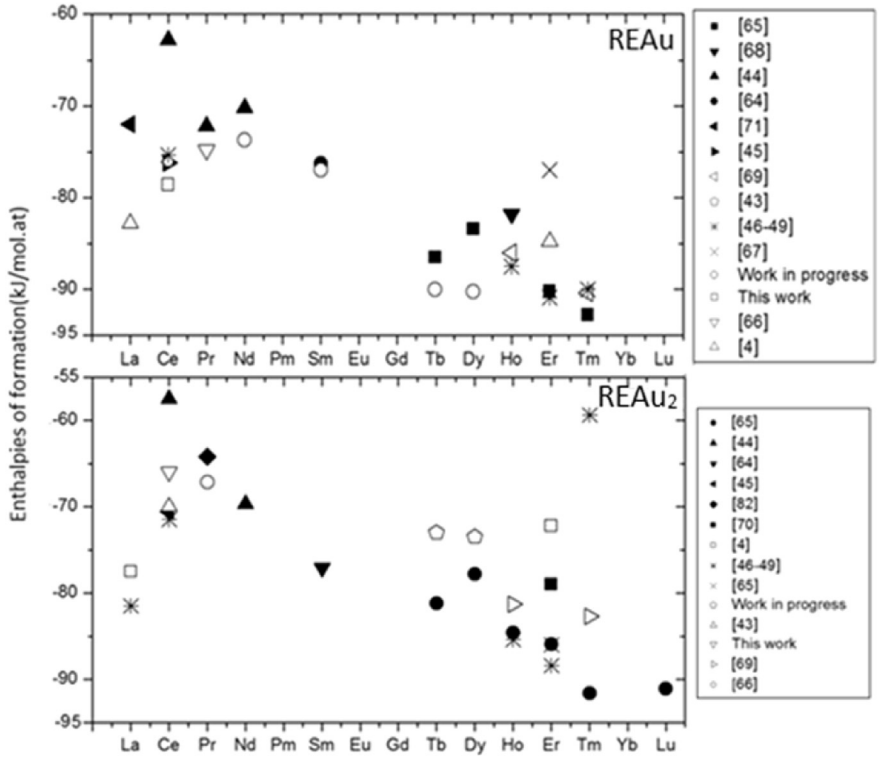

Fig. 8. Enthalpies of formation of the REAu and $REAu_2$ phases with data from work in progress (○and).

and the experimental one from Lomello–Tafin et al. [29] (Fig. 1) is shown in Fig. 2. We note a good agreement between the calculated compositions and temperatures of the relevant invariant reactions and the experimental phase boundaries from Ref. [29] with the exception of the slight deviation of the calculated phase diagram (see Fig. 2) correcting the unusual asymmetry of the liquidus curves around the βCeAu phase which appears in the experimental phase diagram (Fig. 1). Indeed, according to Okamoto and Massalski's studies [60,61], judgment about asymmetry of liquidus curves can be made comparing the ratio of the width of the two phase fields on each side of a compound at the same temperature. If the width on one side of the compound is either one half or double that on the other side, the asymmetry may be considered clearly unusual.

Similar situations were encountered in the literature. We summarize in Table 5, the experimental and calculated data concerning the liquid + REAu ↔ RE_2Au invariant reaction in RE–Au (RE = Ce, Nd, Sm, Tb, Pr, Tm, Er, La, Ho) systems available in the literature. We noted that except for La–Au and Ho–Au systems, all the cited RE–Au experimental diagrams present an asymmetric liquidus around the equiatomic REAu compound. The optimization work by Dong et al. [43] reported similar situations and corrected this anomaly for some studied RE–Au systems.

As mentioned previously, the enthalpies of formation of the Ce–Au intermetallic compounds are measured by several authors (Table 2) by means of direct reaction calorimetry at high temperature. We noted that there is a discrepancy between the standard enthalpies of formation measured by Fitzner and Kleppa [44] and by Janghorban et al. [45]. The difference between the experimental values is of 13, 4 (kJ/mol. at) for the CeAu equiatomic phase and 13,1 (kJ/mol. at) for the $CeAu_2$ compound (Fig. 3). This discrepancy between the experimental results is certainly due to the relative purity of the starting materials and to the fact that the rare earth elements are largely reactive with crucibles and containers, particularly at high temperatures largely developed in the work by Refs. [50,51]. According to the data available in the literature concerning the evolution of the enthalpies of formation of the RE–Au compounds.

Fig. 8, we decided to use the recent experimental of Janghorban et al. [45] as starting values to achieve the present optimization work. Fig. 3 shows the standard enthalpies of formation of Ce–Au intermediate phases together with the calculated values by ab initio method [46–49] as well as optimized ones by Dong et al. [43] and the experimental data available in the literature [39,44,45]. We noticed a good agreement between the starting experimental enthalpies of formation [45] and our calculated values. However, we note that our calculated value for the Ce_2Au phase is slightly more exothermic than the only experimental enthalpy measured by Ferro et al. [39]. On the other hand, our calculated enthalpy of formation agree well with the optimized one (Fig. 3) by Dong et al. [43].

In order to check the reliability of our thermodynamic parameters, we calculated the Ce–Au phase diagram suspending the liquid phase as recommended by some authors [62,63]. Indeed, Fig. 4 shows that the primary bcc_A2 phase is more stable against the defined compounds. This extra criterion provides a better test of the reliability of the Gibbs energy descriptions for extrapolation, especially at high temperatures.

The calculated Gibbs energy of mixing of the liquid phase is also plotted (Fig. 5) at different compositions in the temperatures range 1000 K < T < 4000 K to show that there is no reversed gap in the liquid phase until 4000 K. Finally, as shown in Fig. 6, the assessed enthalpies of mixing of the liquid phase are in good agreement with the experimental data measured by calorimetry at 1473 ± 2 K [44].

In Fig. 7 and 8 we plot the enthalpies of formation of the RE–Au compounds extracted from the literature [64–82], against the atomic number of the rare earth elements (RE). The experimental data are represented as fulfilled forms whereas the values obtained from thermodynamic modelization and theoretical models are represented by empty symbols and asterics respectively. We can

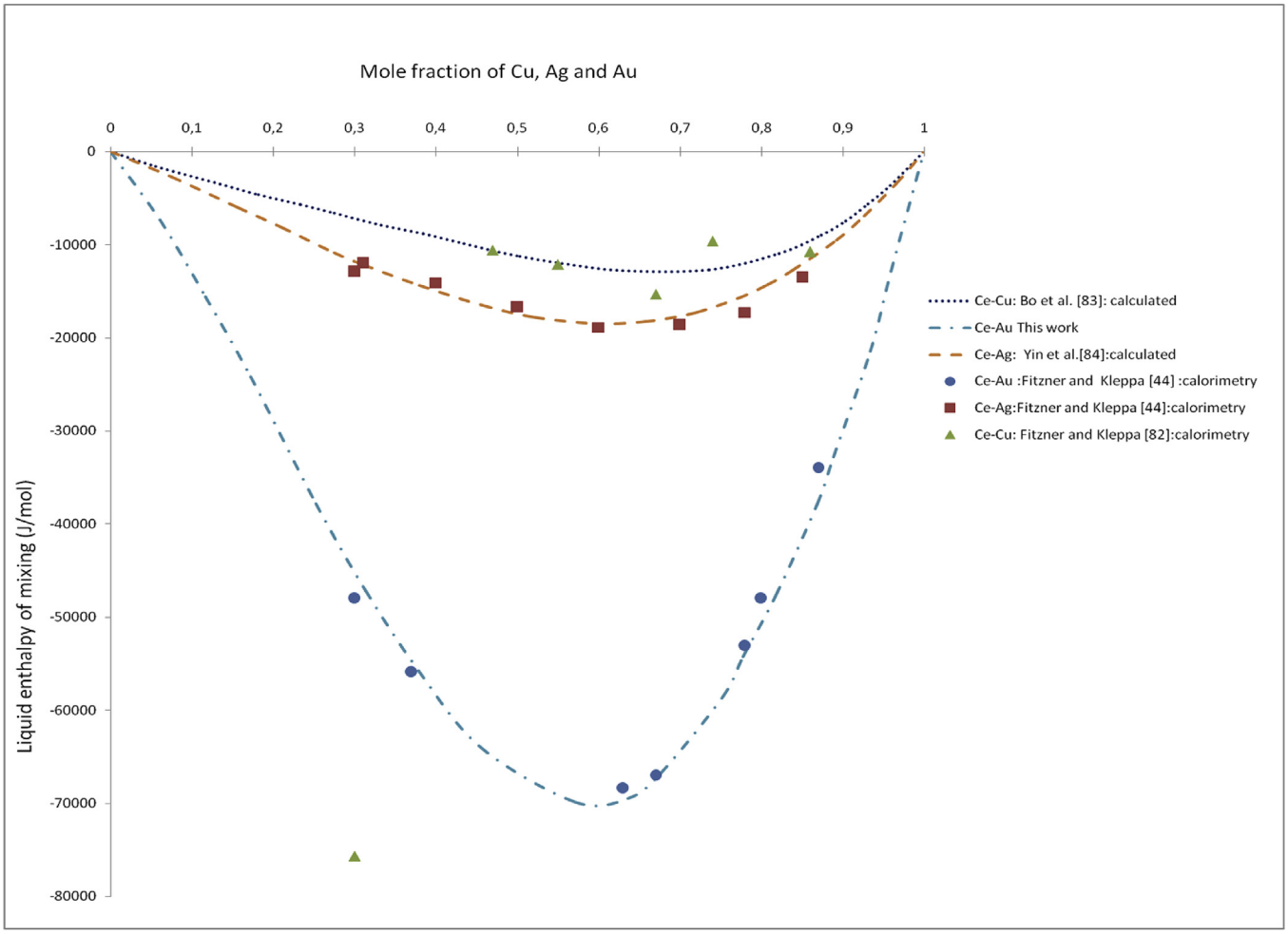

Fig. 9. Calculated enthalpies of mixing of Ce–X (X = Cu, Ag and Au) liquids together with the experimental results from Refs. [44,82].

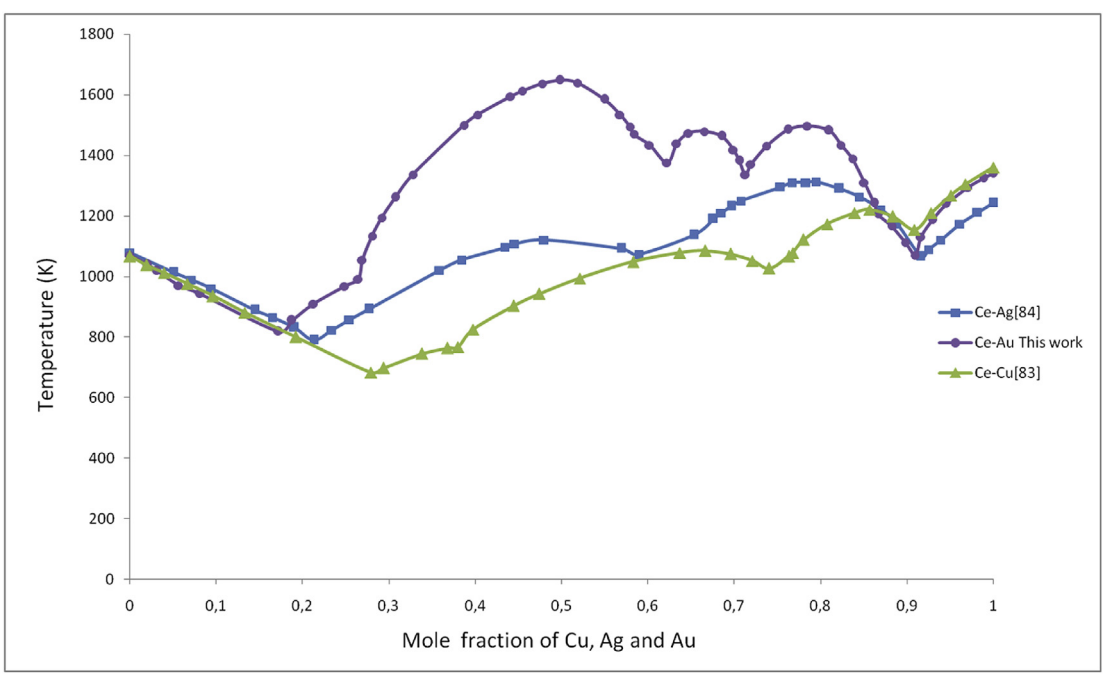

Fig. 10. Liquidus curves in the Ce–X (X = Cu, Ag and Au) binary systems.

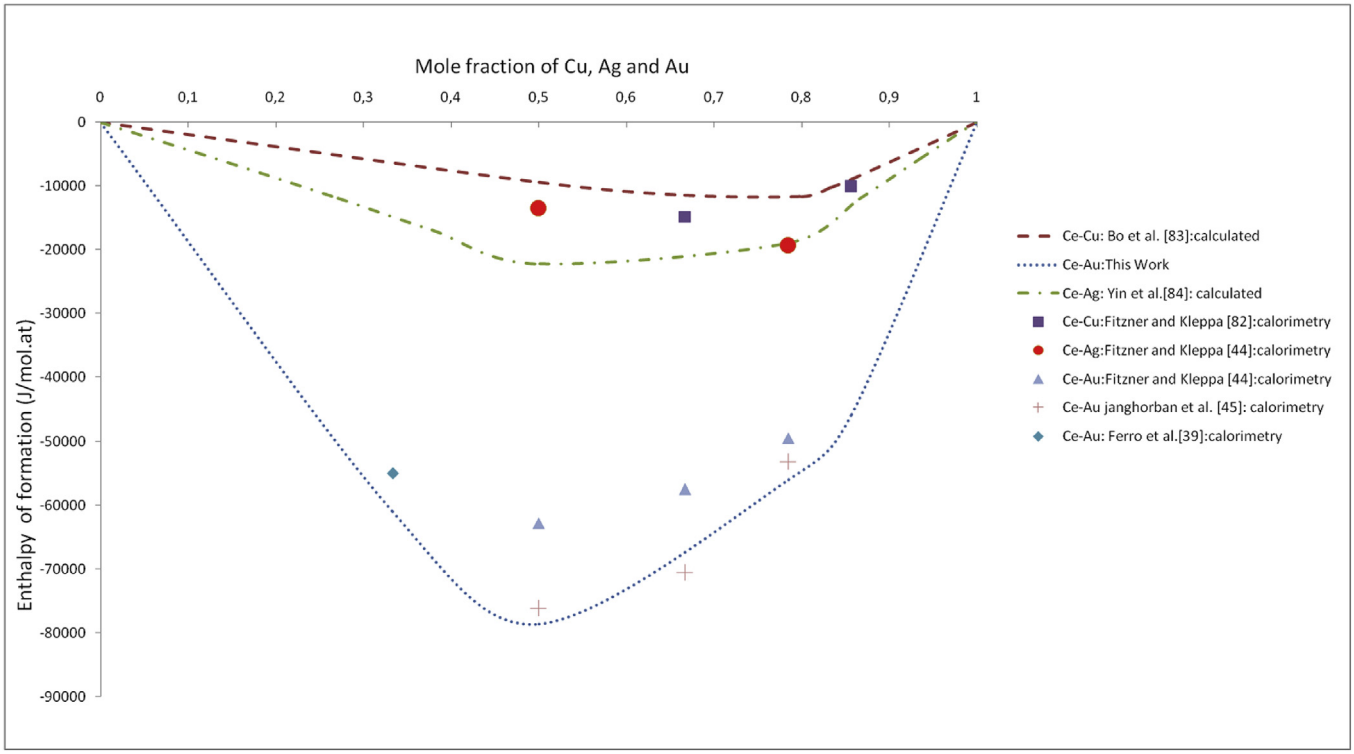

Fig. 11. Calculated standard enthalpies of formation of Ce–X (X = Cu, Ag and Au) intermetallic compounds compared with calorimetric measurements by Refs. [44,45,82].

note that the variation of those enthalpies increase with the atomic number. Our optimized values obtained either for the Ce–Au system or others works in progress agree well with the noted variation of the RE–Au enthalpy of formation.

Fig. 9 presents the change of the enthalpy of mixing the liquid phase measured by Fitzner and Kleppa [44,82] using direct synthesis calorimetry at 1473 ± 2 K for the binary systems Ce–X, where X = Cu, Ag and Au and calculated by several authors [83,84] by means of thermodynamic optimization. As expected, the liquid enthalpy of mixing of the Ce–X liquid alloys becomes more negative in the following order: Cu, Ag and Au. On the other hand, the maximum ordering composition in the liquid solution is located in the vicinity of the molar fraction x = 0.6 of the noble metals. In the course of a study of the trend in the binary systems involving Nd and transition metals from Mn to Cu, Hussain et al. [85] noted that the maximum ordering composition in the liquid solution is shifting toward the transition metal side and stronger compounds in the order: Mn, Fe, Co, Cu and Ni.

On the other hand, as can be seen in Fig. 10, the number of compounds with congruent melting behavior, present in the Ce–X (X = Cu, Ag and Au) phase diagrams, increases in the same order: two in the Ce–Cu and Ce–Ag systems and three in the Ce–Au system. A similar trend was reported by Hussain et al. [85] for the Nd–TM systems (TM = Mn, Fe, Co, Ni and Cu) in the following order: Mn, Fe, Co, Cu and Ni. These authors pointed out that the reason for this trend is not clear and that thermochemical and/or electronic properties of these elements are possibly at the origin of the changes observed in the phase diagrams and thermodynamic properties.

Fig. 11 presents the experimental standard enthalpies of formation of Ce–X (X = Cu, Ag and Au) intermetallic compounds measured by Fitzner and Kleppa [44,82] and by Janghorban et al. [45], using direct reaction calorimetry at high temperature, together with the calculated values from the current work and other optimization studies from literature [83,84]. We note that the enthalpy of formation increase with the atomic number of the X element (X = Cu, Ag and Au). The very strong interaction of Ce with the noble metals is noted for the late element of this group. We observe also, that the shape of the enthalpy of formation versus composition present a pronounced shift toward the equiatomic composition for the Ce–Au alloys. This behavior was already reported by Colinet [86] in an extensive review of the systematic of the enthalpies of formation of rare earths with transition metals. This author mentioned that the shape of the enthalpy of formation of rare earths with late transition metals becomes less asymmetrical from Ni to Pd to Pt. Furthermore, Gschneidner [87,88] proposed that the variation in the enthalpy of formation in a compound series is correlated with the lanthanide contraction in compound series relative to the contraction in the pure metals. This correlation can be used to predict unknown values if at least one enthalpy of formation value is known in the considered series of compounds [86]. As pointed out by Colinet [86], two contradictory effects occur considering the alloys of rare earths: a chemical effect which leads to a very negative contribution to the enthalpy of formation and a volume effect due to size mismatch between the rare earth and the other elements which leads to a positive contribution to the enthalpy of formation.

In the tight binding approach applied to predict the enthalpies of formation of binary intermetallic compounds, Colinet and coworkers [89,90] observed an increase in the enthalpies of formation of a compound of given stoichiometry with the atomic number and tried to obtain a better fit of the experimental values by modifying the hybridization term along the rare–earth series. The fact that the hybridization term varies along the rare earth series was explained by considering the f–p hybridization. This hybridization changes on going from light to heavy rare earth elements and can be related to the participation of the 4f electrons in the bonding. As suggested by Gschneidner and Valletta [91], Colinet et al. [92] considered that a measure of this participation is given by r_R/r_{4f}: the ratio of the

metallic radius of the rare earth metal to 4f radius calculated from the data of Freeman and Watson [93].

6. Conclusion

The experimental phase equilibria and thermodynamic data relative to the Ce–Au system available in the literature have been critically evaluated. Within the scheme of the CALPHAD technique, an appropriate thermodynamic models were used for all the solution phases and intermediate compounds. A two sublattice–phase energy model was employed to describe the substoichiometric CeAu phase. A set of consistent thermodynamic parameters for the Ce–Au binary system is obtained. The calculated phase equilibria and the thermodynamic properties, such enthalpies of mixing the liquid phase, standard enthalpies of formation of the Ce–Au intermediate phases and Gibbs free energies of mixing of the liquid were confronted to the available data from literature. A good agreement was obtained.

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